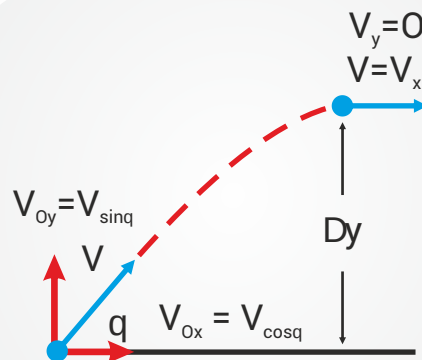


PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Kinematics-2D

(English Medium)

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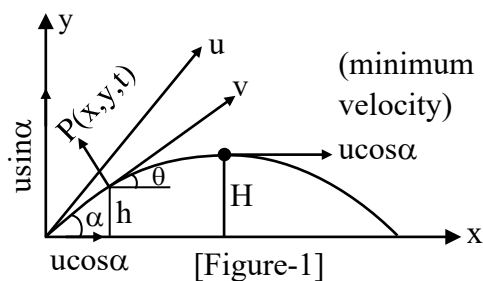
KINEMATICS-2D

KEY CONCEPT

1. GROUND TO GROUND PROJECTILE MOTION :

When a body is thrown obliquely (in a vertical plane) into the uniform gravitational field then the trajectory (actual path of motion) is a parabola. The horizontal component of velocity $u \cos \alpha$ remains unchanged whereas vertical component decreases up to the maximum height and then increases.

- (a) Time taken to reach the height point $t_H = \frac{u \sin \alpha}{g}$



- (b) Maximum height $H = \frac{u^2 \sin^2 \alpha}{2g}$
- (c) Total time of flight $= \frac{2u \sin \alpha}{g} = 2t_H$
- (d) Horizontal range $= (u \cos \alpha) \cdot T = \frac{2}{g} (u \cos \alpha) (u \sin \alpha) = \frac{u^2 \sin 2\alpha}{g}$
- (e) $R_{\max} = \frac{u^2}{g}$ if $\alpha = 45^\circ$

Note that for a given velocity of projection & a given horizontal range there are in general two directions of projection which are complement of each other and are equally inclined to the direction of the maximum range.

(f) **Velocity & Direction Of Motion At A Given Time :**
$$\left. \begin{aligned} v \cos \theta &= u \cos \alpha \\ v \sin \theta &= u \sin \alpha - gt \end{aligned} \right\}$$

Squaring & adding these 2 equations we will get the velocity of the projectile. Dividing the velocities in y and x directions gives the direction of motion.

(g) **Velocity & Direction Of Motion At A Given Height h :**

$$\left. \begin{aligned} v^2 \cos^2 \theta &= u^2 \cos^2 \alpha \\ v^2 \sin^2 \theta &= u^2 \sin^2 \alpha - 2gh \end{aligned} \right\} \text{ on adding } v^2 = u^2 - 2gh$$

(h) **Equations Of Motion In Vector Notation :**

(i) $\vec{v} = \vec{u} + \vec{g}t$ (ii) $\vec{S} = \vec{u}t + \frac{1}{2} \vec{g}t^2$ (iii) $\vec{v}_{av} = \frac{\vec{S}}{t} = \vec{u} + \frac{1}{2} \vec{g}t$ ($\vec{v}_{av}$ = average velocity vector)

(iv) $\vec{v} \cdot \vec{v} = \vec{u} \cdot \vec{u} + 2\vec{a} \cdot \vec{S}$

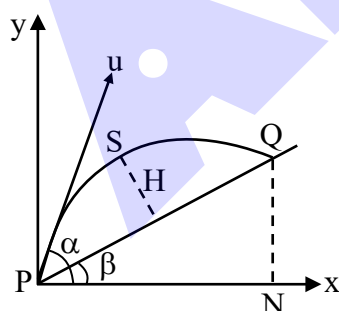
(i) **Equation of Trajectory :**

$$y = x \tan \alpha - \frac{gx^2}{2u^2 \cos^2 \alpha} = x \tan \alpha \left(1 - \frac{x}{R} \right)$$

Note that $\frac{dy}{dx}$ represent the direction of motion.

2. PROJECTILE UP AN INCLINED PLANE ($\alpha \rightarrow$ angle with horizontal):

(a) Total time of flight on the inclined plane $T = \frac{2u \sin(\alpha - \beta)}{g \cos \beta}$



(b) Range PQ on the inclined plane

$$PQ = \frac{2u^2 \cos \alpha \cdot \sin(\alpha - \beta)}{g \cos^2 \beta} = \frac{u^2}{g \cos^2 \beta} [\sin(2\alpha - \beta) - \sin \beta]$$

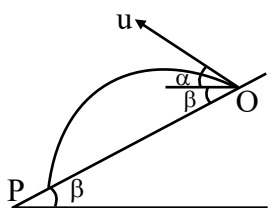
(c) For Maximum range $2\alpha - \beta = \frac{\pi}{2} \Rightarrow \alpha = \frac{\pi}{4} + \frac{\beta}{2}$

Hence the direction for maximum range bisects the angle between the vertical and the inclined plane.

(d) Greatest distance of the projectile from the inclined plane ; $H = \frac{u^2 \sin^2 (\alpha - \beta)}{2g \cos \beta}$ when the projectile is at S, its velocity perpendicular to the plane is zero.

3. PROJECTILE DOWN AN INCLINED PLANE :

(a) Time of flight = $\frac{2u \sin (\alpha + \beta)}{g \cos \beta}$



(b) Range OP = $\frac{2u^2 \sin (\alpha + \beta) \cdot \cos \alpha}{g \cos^2 \beta}$

(c) Angle of projection α for maximum range = $\frac{\pi}{4} - \frac{\beta}{2}$

4. RELATIVE VELOCITY :

Velocity of 'A' relative to 'B' is given by $\vec{V}_{AB} = \vec{V}_A - \vec{V}_B$.

$\vec{V}_{AB}$ refers to the velocity which 'A' appears to have as seen by B. The above idea of 1 dimensional relative motion can be extended to motion in 2 dimensions.

- For crossing a river in minimum possible time, the velocity of the swimmer relative to the river should be perpendicular to the river flow.
- For crossing a river travelling least possible distance, the velocity of the swimmer relative to the ground should be as close to the perpendicular to the river flow as possible.

- (c) If an athlete is running, his clothes will be furling in the direction of the velocity of the wind relative to him self.
- (d) If you do not want to get drenched in a rain, you should hold the umbrella in a direction opposite to the velocity of the rain relative to your self.
- (e) Two particles, 1 and 2, move with constant velocities v_1 and v_2 . At the initial moment their radius vectors are equal to r_1 and r_2 . For the two particles to collide, the relative velocity of 1 with respect to 2 should be directed towards 2 as seen by 2.
- (f) If the 2 particles do not collide, at the minimum separation the relative velocity should be perpendicular to line joining both.